② TF - IDF [Term Frequency - Inverse Document Frequency]

S1 → good boy
S2 → good girl
S3 → boy girl good

$$\text{Term Freq (TF)} = \frac{\text{No. of rep of words in sentence}}{\text{No. of word in sentence}}$$

$$\text{IDF} = \log_e \left(\frac{\text{No. of sentence}}{\text{No. of sentences contain the word}} \right)$$

Term Frequency_ ---> * <---

	S1	S2	S3
good	1/2	1/2	1/3
boy	1/2	0	1/3
girl	0	1/2	1/3

Note: In TF means how many time that word present in that sentence. (particular sentence)

In IDF means how many time the word present in all the sentence.

IDF

Word	IDF
good	$\log_e(3/3) = 0$
boy	$\log_e(3/2)$
girl	$\log_e(3/2)$

Multiply → TF * IDF

Final TF - IDF

o/p

	[good	boy	girl]
sent 1	0	$1/2 \times \log_e(3/2)$	0
sent 2	0	0	$1/2 \times \log_e(3/2)$
sent 3	0	$1/3 \times \log_e(3/2)$	$1/3 \times \log_e(3/2)$

BOW

1	1	0
1	0	1
1	1	1